

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
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Section / Batch:	BE-CSE	Date:	18/07/2026

Instructions to Students

- Answer the following question. It carries 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1

Q1. Genome Sequencing Analysis

Given Recurrence

$$T(n) = 3T(n/2) + n$$

We will apply the Master Theorem to determine the time complexity.

Step 1: Identify the Parameters

The standard form of the Master Theorem is:

$$T(n) = aT(n/b) + f(n)$$

Comparing with the given recurrence:

- $a = 3$ (Number of recursive calls)
 - $b = 2$ (Input size reduced by half)
 - $f(n) = n$
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Step 2: Calculate $n^{\log_b a}$

$$\log_b a = \log_2 3 \approx 1.585$$

Therefore,

$$n^{\log_2 3} = n^{1.585}$$

Step 3: Compare $f(n)$ **with** $n^{\log_b a}$

We have

$$f(n) = n = n^1$$

Since

$$n < n^{1.585},$$

$$f(n) = O(n^{1.585-\varepsilon})$$

where

$$\varepsilon = 1.585 - 1 = 0.585 > 0$$

Thus, $f(n)$ **is polynomially smaller than** $n^{\log_b a}$.

Step 4: Identify the Master Theorem Case

This satisfies Case 1 of the Master Theorem:

If

$$f(n) = O(n^{\log_b a - \varepsilon})$$

for some $\varepsilon > 0$, then

$$T(n) = \Theta(n^{\log_b a})$$

Step 5: Final Time Complexity

$$T(n) = \Theta(n^{\log_2 3})$$

or approximately

$$T(n) = \Theta(n^{1.585})$$

Justification

- The algorithm divides the genome into 3 recursive subproblems, each of size $n/2$.
- The additional work outside recursion is linear ($O(n)$).
- Since the recursive work grows faster than the linear work, the recursion dominates the overall running time.
- Hence, according to Master Theorem Case 1, the time complexity is:

$$\Theta(n^{1.585})$$

Scalability Analysis for Large Biological Datasets

Genome sequencing involves processing extremely large DNA datasets, often containing millions or billions of base pairs.

Advantages

- The algorithm has a complexity of $\Theta(n^{1.585})$, which is better than quadratic time $O(n^2)$.
- Dividing the genome into smaller subproblems enables efficient processing of large datasets.
- Recursive decomposition makes the algorithm suitable for parallel processing, where multiple processors can work on different genome segments simultaneously.

Limitations

- The running time still increases faster than linear time as dataset size grows.
- Very large genomes require additional memory for recursive calls.
- Processing whole-genome datasets may become computationally expensive without high-performance computing resources.

Scalability Conclusion

The algorithm scales reasonably well for large biological datasets because its growth rate is sub-quadratic. Although execution time increases with genome size, it remains significantly more efficient than $O(n^2)$ algorithms. Performance can be further improved using parallel computing, making this approach practical for modern genome sequencing applications.

Final Answer

- Parameters: $a = 3, b = 2, f(n) = n$
- Computed Value: $n^{\log_2 3} = n^{1.585}$
- Master Theorem Case: Case 1
- Time Complexity: $\Theta(n^{1.585})$
- Scalability: Efficient for large genome datasets due to sub-quadratic complexity and good potential for parallel execution, though computational cost still increases as dataset size becomes very large.

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I Sairaj C (192411018) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sairaj C

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach

Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____